Name: _____

Student Workbook

Fall, 2022 - Codap Edition

Workbook v1.5

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Introduction to Computational Data Science

Many important questions ("What's the best restaurant in town?", "Is this law good for citizens?", etc.) are answered with *data*. Data Scientists try and answer these questions by writing *programs that ask questions about data*.

Data of all types can be organized into **Tables**.

- Every Table has a **header row** and some number of **data rows**.
- **Quantitative data** is numeric and measures *an amount*, such as a person's height, a score on a test, distance, etc. A list of quantitative data can be ordered from smallest to largest.
- **Categorical data** is data that specifies *qualities*, such as sex, eye color, country of origin, etc. Categorical data is not subject to the laws of arithmetic — for example, we cannot take the "average" of a list of colors.

The Animals Dataset

name	species	sex	age	fixed	legs	pounds	weeks
Sasha	cat	female	1	false	4	6.5	3
Snuffles	rabbit	female	3	true	4	3.5	8
Mittens	cat	female	2	true	4	7.4	1
Sunflower	cat	female	5	true	4	8.1	6
Felix	cat	male	16	true	4	9.2	5
Sheba	cat	female	7	true	4	8.4	6
Billie	snail	hermaphrodite	0.5	false	0	0.1	3
Snowcone	cat	female	2	true	4	6.5	5
Wade	cat	male	1	false	4	3.2	1
Hercules	cat	male	3	false	4	13.4	2
Toggle	dog	female	3	true	4	48	1
Boo-boo	dog	male	11	true	4	123	24
Fritz	dog	male	4	true	4	92	3
Midnight	dog	female	5	false	4	112	4
Rex	dog	male	1	false	4	28.9	9
Gir	dog	male	8	false	4	88	5
Max	dog	male	3	false	4	52.8	8
Nori	dog	female	3	true	4	35.3	1
Mr. Peanutbutter	dog	male	10	false	4	161	6
Lucky	dog	male	3	true	3	45.4	9
Kujo	dog	male	8	false	4	172	30
Buddy	lizard	male	2	false	4	0.3	3
Gila	lizard	female	3	true	4	1.2	4
Bo	dog	male	8	true	4	76.1	10
Nibblet	rabbit	male	6	false	4	4.3	2
Snuggles	tarantula	female	2	false	8	0.1	1
Daisy	dog	female	5	true	4	68	8
Ada	dog	female	2	true	4	32	3
Miaulis	cat	male	7	false	4	8.8	4
Heathcliff	cat	male	1	true	4	2.1	2
Tinkles	cat	female	1	true	4	1.7	3
Maple	dog	female	3	true	4	51.6	4

Categorical or Quantitative?

- **Quantitative data** measures an *amount* and can be ordered from smallest to largest.
- **Categorical data** specifies *qualities* and is not subject to the laws of arithmetic — for example, we cannot take the "average" of a list of colors.

Note: Numbers can be sometimes be categorical rather than quantitative!

For each piece of data below, circle whether it is **Categorical** or **Quantitative** data.

1)	Hair color	categorical	quantitative
2)	Age	categorical	quantitative
3)	ZIP Code	categorical	quantitative
4)	Year	categorical	quantitative
5)	Height	categorical	quantitative
6)	Sex	categorical	quantitative
7)	Street Name	categorical	quantitative

For each question, circle whether it will be answered by **Categorical** or **Quantitative** data.

8)	We'd like to find out the average price of cars in a lot.	categorical	quantitative
9)	We'd like to find out the most popular color for cars.	categorical	quantitative
10)	We'd like to find out which puppy is the youngest.	categorical	quantitative
11)	We'd like to find out which cats have been fixed.	categorical	quantitative
12)	We want to know which people have a ZIP code of 02907.	categorical	quantitative
13)	We'd like to sort a list of phone numbers by area code.	categorical	quantitative

Questions and Column Descriptions

What questions can you ask about the animals dataset? For each question, **can it be answered by this dataset?** Make sure you have at least two questions that can be answered, and at least one that cannot.

What do you NOTICE about this dataset?	What do you WONDER about this dataset?	Answered by this dataset?
		Yes *No*
		Yes *No*
		Yes *No*
		Yes *No*
		Yes *No*
		Yes *No*
		Yes *No*

Describe the table, and two of the columns, by filling in the blanks below.

1. This dataset is _____, which contains _____ data rows.
2. Some of the columns are:
   a. _____ (column name), which contains _____ (*categorical or quantitative*) data. Some example values are:
   _____.
   b. _____ (column name), which contains _____ (*categorical or quantitative*) data. Some example values are:
   _____.

What's on your mind?

Opening Questions

Sports

- Who is the best quarterback of all time?
- Are baseball pitchers throwing harder than ever?
- How much more do male soccer players earn than females?
- How common is it for former Olympic athletes to become coaches?
- How much does an extra inch of height help a basketball player?

Pop Culture

- What percentage of people have seen the movie that won last year's Best Picture Award?
- Who tends to be more popular: bands or solo singers?
- Are younger actors paid more than older actors?
- Are movies with female leads as profitable as movies with male leads?
- Does winning a Grammy increase sales?

Politics

- Is "Stop and Frisk" a racist policy?
- Do Republican politicians tend to come from different states than Democratic ones?
- Do people in countries that have universal healthcare live longer than people in countries that don't?
- Was press coverage slanted for or against a particular candidate?

Education

- Do small schools perform better than large ones?
- Which has a stronger correlation with student achievement: race or wealth?
- Do bilingual classes result in better outcomes for ESL/ELL students?

What Questions Can You Answer with the Given Data?

The following is a dataset of a bicycle rider's training rides.

date	miles	time (w/stops)	weather	average speed	max speed
04/10/2018	10	44	"cloudy"	13	30
05/30/2018	15	66	"sunny"	13.5	22
06/12/2018	12	61	"rainy"	11.2	25
07/04/2018	24	103	"sunny"	14	26
07/12/2018	24	120	"windy"	12.5	26

1) Decide whether each questions below *can* or *cannot* be answered with the given data and circle your selection.

2) In the space provided below, explain *how* you could answer the question using the data or *why you cannot* answer the question.

Question	Answered by this dataset?
How many miles did the cyclist ride June 12th?	*Yes _No*
What tire pressure produces the highest average speed?	*Yes* _No
What is the average time it takes this cyclist to ride 1 mi?	*Yes _No*
Does this cyclist ride slower when it is rainy?	*Yes* _No
Does this cyclist ride faster when they are late to an appointment?	*Yes* _No
How many miles has the cyclist ridden in total as part of their training?	*Yes _No*

Are there any questions that you could find the answers to more than one way?

Simple Data Types

CODAP is a web-based data science tool that runs in your web browser.

Data Types

CODAP utilizes different ***data types***, including Numbers, Strings, and Booleans.

- Numbers are values like `1`, `0.4`, `1/3`, and `-8261.003`.
  - Numbers are *usually* used for quantitative data and other values are *usually* used as categorical data.
- Strings are values like `"Emma"`, `"Rosanna"`, `"Jen and Ed"`, or even `"08/28/1980"`.
  - All strings *must* be surrounded in quotation marks.
- Booleans are either `true` or `false`.

All values evaluate to themselves. The program `42` will evaluate to `42`, the String `"Hello"` will evaluate to `"Hello"`, and the Boolean `false` will evaluate to `false`.

Operators

Operators (like `+`, `-`, `<`, etc.) work the same way that they do in math.

- Operators are written between values, for example: `4 + 2`.

Expressions

Expressions work the same way that they do in math. Numeric expressions can be *evaluated*.

- The following are all examples of expressions: `sqrt(16)`, `sqrt(Weight)`, `m+5`, and `9+17`.

CODAP Exploration

Make sure you're logged into the Animals Starter File in CODAP. The purpose of this exploration is to familiarize yourself with some of CODAP's features. Feel free to experiment and try things out!

1) See if you can make a graph. What did you select in order to do so?

2) Move the table. Move the graph. How do you move things in CODAP?

3) Minimize and maximize the table and the graph. How do you do that?

4) Drag an attribute (like `weight`, `name`, or `sex`) to a graph's axis - or use the drop-down menu that appears when you click on an axis. What happens?

5) Drag a second attribute (like `weight`, `name`, or `sex`) to the graph's other axis - or use the drop-down menu that appears when you click on an axis. What happens?

6) Select the grey plus sign (+) in the upper right-hand corner of the Animals Table. Doing so will enable you to create a new attribute. Name your new attribute `test`. Is the column populated or empty?

Numbers and Strings

Make sure you're logged into CODAP, and have created a new column titled `test` in the Animals Dataset. In order to follow each of these directives, you must first select `test`, and then choose `Edit formula` from the drop-down menu that appears. You will be typing into the `Edit Formula` box.

Numbers

1) Try typing `42` and then select `Apply`. What happens? What is the largest number CODAP can handle?

2) What happens if you try a fraction like `1/5`? How about `1/3`?

3) Try writing negative integers, fractions and decimals.

Strings

String values are always in quotes.

4) Try typing your name in the `Formula` box. What error message appears? Why do you think it appears?

5) Try typing your name in quotes.

6) Try typing your name with the opening quote, but *without the closing quote*. Read the error message!

7) Is `42` the same as `"42"`? Why or why not? Write your answer below:

8) Explain what you understand about how strings work in CODAP.

Operators

9) Just like math, CODAP has ***operators*** like `+`, `-`, `*` and `/` and `^`. Try typing in `4 + 2`.

10) Type in and apply the following expressions, **one at a time**: `4 + 2 * 6`, `(4 + 2) * 6`, `4 + (2 * 6)`. What do you notice? Does CODAP know the order of operations?

11) Try typing in `4 + "cat"`, and then `"dog" + "cat"`. What can you conclude from this?

Booleans

Boolean-producing expressions are yes-or-no questions and will always evaluate to either `true` ("yes") or `false` ("no"). What will each of the expressions below evaluate to? **In order to try out the code, you must first select `test`, and then choose `Edit formula` from the drop-down menu that appears.** You will be typing into the `Edit Formula` box.

	Prediction:	Computer Returns:		Prediction:	Computer Returns:
1) `3 <= 4`	_____	_____	2) `"a" > "b"`	_____	_____
3) `3 = 2`	_____	_____	4) `"a" < "b"`	_____	_____
5) `2 < 4`	_____	_____	6) `"a" = "b"`	_____	_____
7) `5 >= 5`	_____	_____	8) `"a" != "a"`	_____	_____
9) `4 >= 6`	_____	_____	10) `"a" >= "a"`	_____	_____
11) `3 != 3`	_____	_____	12) `"a" != "b"`	_____	_____

13) In your own words, describe what `<` does. _____

14) In your own words, describe what `>=` does. _____

15) In your own words, describe what `!=` does. _____

	Prediction:	Computer Returns:
16) `includes("cat", "catnap")`	_____	_____
17) `includes("catnap", "cat")`	_____	_____

18) How many **Numbers** are there in the entire universe? _____

19) How many **Strings** are there in the entire universe? _____

20) How many **Booleans** are there in the entire universe? _____

Expressions & Functions in CODAP

Complete the tables below.

$y = 3x$			$y = x + 2$	
x	y		x	y
5			3	
3.5			8	
7.1			1	
8.4			6	
9.2			5	

1) Open your Animals Starter File. Edit the `test` column's formula to compute *triple* the weight of all of the animals. What formula will you use? *Note: Attribute names that are more than one word need to be entered inside of tick marks.*

_____

- Compare your results with the first table, above.

2) Edit the `test` column's formula so that 2 weeks are added to the time to adoption for all of the animals. What formula did you use?

_____

Applying Functions

In order to answer each of the questions below, you must first select `test` in Animals Starter File, and then choose `Edit formula` from the drop-down menu that appears. You will be typing into the `Edit Formula` box.

- Enter `sqrt(16)` into the `Edit Formula` box.

3) How many arguments does `sqrt` expect? _____

4) What type of argument does the function expect? (Number? String? Boolean?) _____

5) What type of output does `sqrt` produce? _____

6) What did the function evaluate to? _____

- Now, enter `sqrt(Weight)`.

7) What do you observe? _____

8) What happens when you enter `sqrt(Legs)`? _____

9) What about `sqrt(Name)`? _____

- Enter `stringLength(Name)` into the `test` column.

10) What do you observe? _____

11) How many arguments does `stringLength` expect? _____

12) What type of argument does `stringLength` expect? _____

13) Does `stringLength` produce a Number? String? Boolean? _____

Putting it all together

- Create a column called `age-next-year` that uses an expression to add one year to each animal's age.
- Create a column called `rounded-weight` that uses a function to give the weight of each animal rounded to the nearest whole number.
- Create a column called `old` that uses a Boolean to indicate which animals are over 5 years old.

Displaying Categorical Data

Displaying Categorical Variables

- With a table open in CODAP, select the "graph" icon to produce a scatter plot of *randomly distributed* data points.
- Drag attributes/columns to the axes (or select from a drop-down menu of attributes/columns by clicking the axes) to organize the data so that it is no longer randomly distributed.
- Once the data is organized, manipulate it further by selecting the graph menu icons:
  - **the ruler icon** provides options for calculating statistics such as mean, median, and standard deviation
  - for *some* datasets (those with strong correlations), the ruler icon will offer *additional* statistical computations (such as a least squares line or regression line)
  - the **bar graph icon** allows new configurations of the data. Select this option to group data points into bins or create a bar for each point. Clicking on the bar graph icon a second time (for instance, after data is grouped into bins) allows the creation of a histogram (by fusing the dots into bars).

Exploring other Displays

Data Scientists use **displays** to visualize data. You've probably seen some of these charts, graphs and plots yourselves! When it comes to displaying **Categorical Data**, we often rely on **bar charts** and **pie charts**. CODAP, however, only has the capacity to create **bar charts**.

When we want to create a data display in CODAP, it is important to consider the following: Which attributes on which axes? What type of data? What configuration?

Bar charts show the *count or percentage* of rows in each category.

- Bar charts provide a visual representation of the frequency of values in a categorical column.
- Bar charts have a bar for every category in a column.
- The more rows in a category, the taller the bar.
- Bars in a bar chart can be show in *any order*, without changing the meaning of the chart. However, bars are usually shown in some sensible order (bars for the number of orders for different t-shirt sizes might be presented in order of smallest to largest shirt).

Data Displays in CODAP

Make sure you're logged into CODAP, with the Animals Dataset open. Create a graph of randomly generated points by selecting the `Graph` icon, and then respond to the following prompts.

Creating Bar Charts in CODAP

1) Select the y-axis on your graph (where it says "Click here"). On the drop-down menu that apperas, select `fixed`. (*If you prefer, you may also drag the attribute name from the table to the y-axis.*) What do you notice?

2) Now select the x-axis on your graph and select `fixed`. How does the graph change?

3) Select the `configuration` icon (which looks like a bar graph) to the right of the data display. Select `fuse dots into bars`

4) Click the `ruler` icon, and test out each of the two options available (count and percentage). What happens?

5) Now, make a bar chart showing how many animals there are of each species by changing the variable on the x-axis to `species`. Can you figure out how to reconfigure the bar chart as a dot plot?

Numeric versus Categorical Displays

6) Create a graph with `weight` on the x-axis. Then, click on `weight` so that a drop-down menu appears. From this drop-down menu, choose `Treat` **`as`** `Categorical`, and observe how the graph changes. What is going on?

7) Can you `fuse dots into bars` when treating `weight` categorically? Numerically? Why does CODAP *sometimes* invite you to `fuse dots into bars`, but other times the option does not appear?

Practice Plotting

Fill in the tables, predict what the graph will look like (a rough sketch is fine!), and then use CODAP to make the following displays.

1) A bar chart showing the `sex` of animals from the shelter.

Column / Attribute	Type of Data	Configuration
_____ [column used as x-axis] _____ [column used as y-axis]	❑ Quantitative ❑ Categorical	❑ Points ❑ Fuse dots into bars ❑ Bar for each point ❑ Group into bins

Sketch the chart below:	What do you think the display tells us?

2) A bar chart showing the `species` of animals from the shelter.

Column / Attribute	Type of Data	Configuration
_____ [column used as x-axis] _____ [column used as y-axis]	❑ Quantitative ❑ Categorical	❑ Points ❑ Fuse dots into bars ❑ Bar for each point ❑ Group into bins

Sketch the chart below:	What do you think the display tells us?

Practice Plotting 2

Fill in the tables, predict what the graph will look like (a rough sketch is fine!), and then use CODAP to make the following displays.

1) A histogram of the number of `pounds` that animals weigh.

Column / Attribute	Type of Data	Configuration
[column used as x-axis] [column used as y-axis]	❑ Quantitative ❑ Categorical	❑ Points ❑ Fuse dots into bars ❑ Bar for each point ❑ Group into bins

Sketch the chart below:	What do you think the data display tells us?

2) A scatter-plot, using the animals `name` as the labels, `age` as the x-axis, and `pounds` as the y-axis, for all the animals from the shelter.

Column / Attribute	Type of Data	Measure
[column used as x-axis] [column used as y-axis] [(optional) column used for labels]	❑ Quantitative ❑ Categorical	❑ Box plot ❑ Least squares line ❑ No need to make a selection

Sketch the chart below:	What do you think the data display tells us?

Practice Plotting - Challenge

Fill in the tables, predict what the graph will look like (a rough sketch is fine!), and then use CODAP to make the following displays.

1) A scatter-plot, using the animals `species` as the labels, `age` as the x-axis, and `weeks` as the y-axis, for all the animals from the shelter.

Column / Attribute	Type of Data	Measure
	❑ Quantitative ❑ Categorical	❑ Box plot ❑ Least squares line ❑ No need to make a selection

Sketch the chart below:	What do you think the data display tells us?

2) (Challenge) A least squares line, using the animals `species` as the labels, `pounds` as the x-axis, and `weeks` as the y-axis, for all the animals from the shelter.

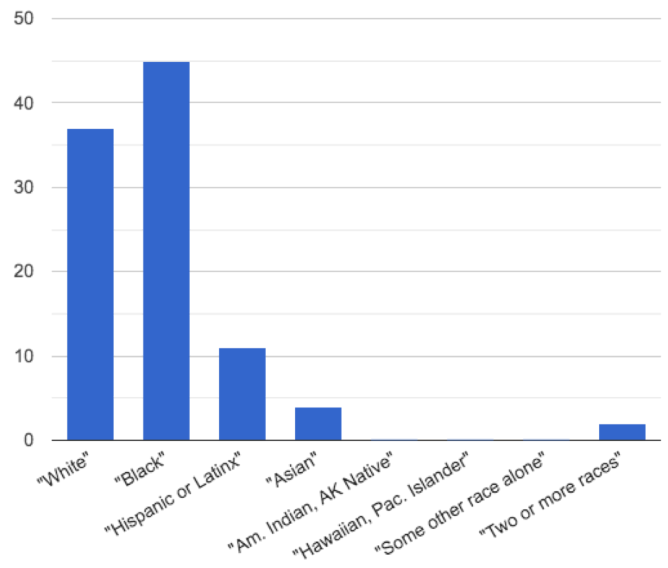
Column / Attribute	Type of Data	Measure
[column used as x-axis] [column used as y-axis] [(optional) column used for labels]	❑ Quantitative ❑ Categorical	❑ Box plot ❑ Least squares line ❑ No need to make a selection

Sketch the chart below:	What do you think the data display tells us?

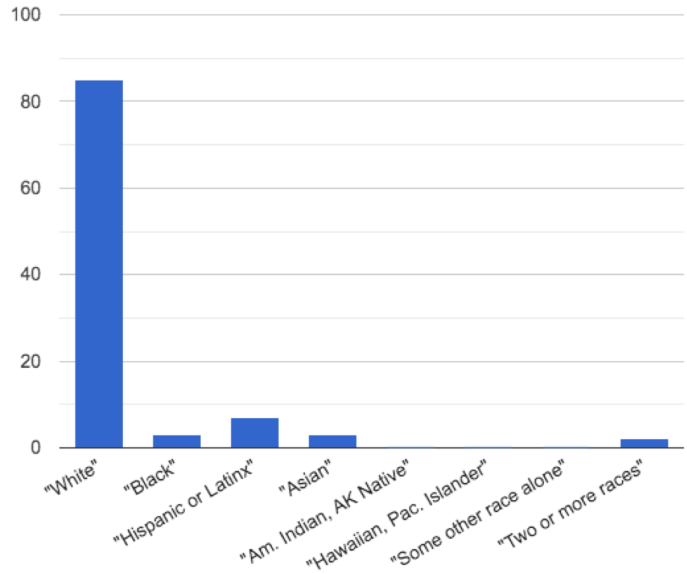
What's on your mind?

Bar Chart - Notice and Wonder

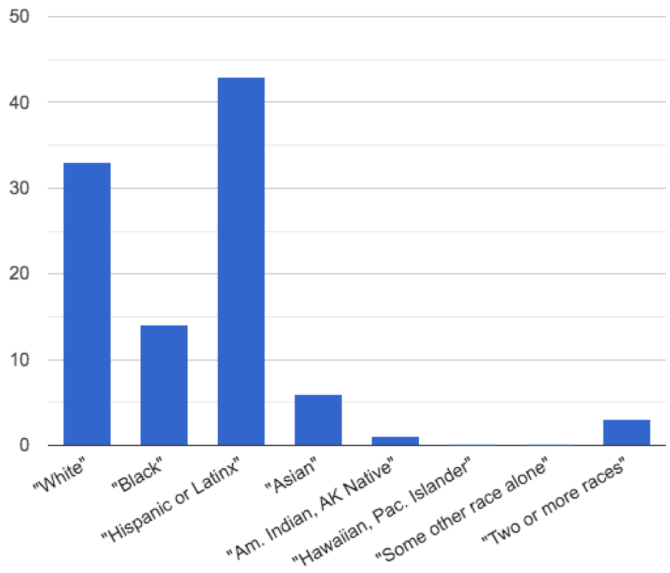
Washington, D.C. Public Schools



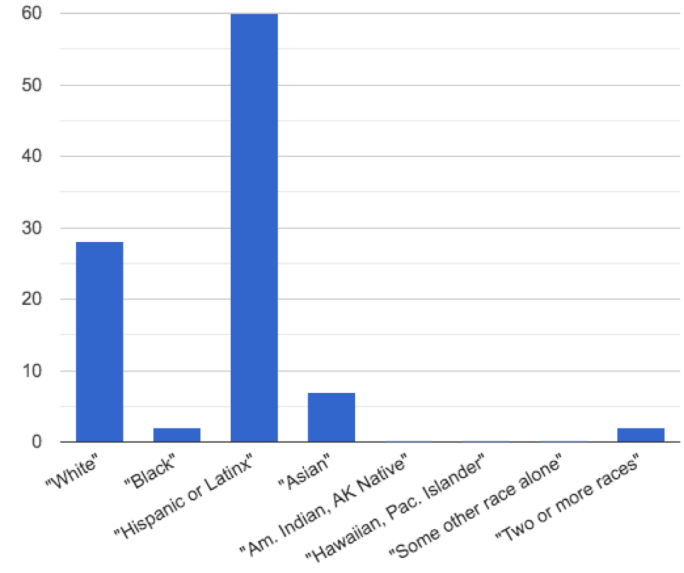
West Warwick Schools, RI



Providence Schools, RI



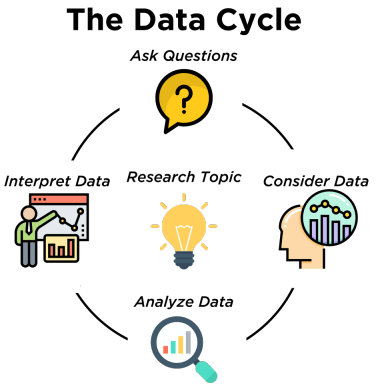
Oxnard Union HS District, CA



What do you Notice?	What do you Wonder?

The Data Cycle

Data Science is all about *asking questions of data*. Sometimes the answer is easy to compute. Sometimes the answer to a question is *already in the dataset* - no computation needed. And sometimes the answer just sparks more questions! Data Scientists ask a lot of questions, and each one adds a chapter to the story of their research. Even if a question is a "dead-end", it's valuable to share what the question was and what work you did to answer it!



We start by **Asking Questions** - questions that can be answered with data. It's useful to think of all questions as falling into four categories:

- **Lookup questions** can be answered simply by looking up a single value in the table. Examples of lookup questions might be "How many legs does Felix have?" or "What species is Sheba?"
- **Arithmetic questions** can be answered by computing an answer across a single column. Examples of arithemetic questions might be "How much does the heaviest animal weigh?" or "What is the average age of animals from the shelter?"
- **Statistical questions** are where things get interesting! The answers are never black-and-white. If we asked, "How old are animals at the shelter?" we could report back the average age, the age that shows up most frequently or the range of the ages. There are also some statistical questions that deal with *relationships between* ***two*** *columns*: "Do cats tend to be adopted faster than dogs?" or "Are older animals heavier than young ones?"
- **Questions we can't answer** are pretty common, too! We might wonder where the animal shelter is located, or what time of year the data was gathered! But the data in the table won't help us answer that question, so as Data Scientists we might need to do some research *beyond the data*. And if nothing turns up, we simply recognize that there are limits to what we can analyze.

Next, we **Consider Data**. This could be done by conducting a survey, observing and recording data, or finding a dataset that meets our needs. Since our data is contained in a table, it's useful to start by asking two questions:

- What rows do we care about? - Is it all the animals? Just the lizards?
- What columns do we need? - Are we examining the ages of the animals? Their weights?

Then, we **Analyze the Data**, by producing data displays and new tables of filtered or transformed data and using them to identify patterns and relationships.

- Are we making a pie chart? A bar chart? Something else?

Finally, we **Interpret the Data**, by answering our questions and summarizing the results. As we've already seen from the Animals Dataset, these interpretations often lead to *new* questions… and the cycle begins again.

Which Question Type?

Take a moment to look at the table of Pokemon data below:

name	type1	hitpoint	attack	defense	speed
Bulbasaur	Grass	45	49	49	45
Ivysaur	Grass	60	62	63	60
Venusaur	Grass	80	82	83	80
Mega Venusaur	Grass	80	100	123	80
Charmander	Fire	39	52	43	65
Charmeleon	Fire	58	64	58	80
Charizard	Fire	78	84	78	100
Mega Charizard X	Fire	78	130	111	100
Mega Charizard Y	Fire	78	104	78	100
Squirtle	Water	44	48	65	43
Wartortle	Water	59	63	80	58

For each question about this data below, write down whether it's a Lookup, Arithmetic, or Statistical Question.

	Question	Type	Which Rows?	Column(s)?
1	What type is Charizard?			
2	Which Pokemon is the fastest?			
3	What is Wartortle's attack score?			
4	What is the average defense score?			
5	What is a typical defense score?			
6	Is Ivy faster than Venusaur?			
7	Does speed affect attack score?			
8	What is the most common type?			
9	Does one type tend to be faster than others?			
10	Are hit-points (hp) similar for all Pokemon?			
11	Which Pokemon has a speed of 43?			

Data Cycle: Consider Data

For each question about the animals dataset below, identify the type of question and fill in the Rows and Columns needed to answer it.

Ask Questions

How old is Boo-boo?

What question do you have?

What type of question is this? (circle one) Lookup Arithmetic Statistical

Consider Data

What Rows should we investigate? (All the rows, just the cats, fixed dogs, etc.)

What Columns do we need? (age, weight-in-kilograms, weeks, etc.)

Ask Questions

Are there more cats than dogs in the shelter?

What question do you have?

What type of question is this? (circle one) Lookup Arithmetic Statistical

Consider Data

What Rows should we investigate? (All the rows, just the cats, fixed dogs, etc.)

What Columns do we need? (age, weight-in-kilograms, weeks, etc.)

Ask Questions

What question do you have?

What type of question is this? (circle one) Lookup Arithmetic Statistical

Consider Data

What Rows should we investigate? (All the rows, just the cats, fixed dogs, etc.)

What Columns do we need? (age, weight-in-kilograms, weeks, etc.)

Ask Questions

What question do you have?

What type of question is this? (circle one) Lookup Arithmetic Statistical

Consider Data

What Rows should we investigate? (All the rows, just the cats, fixed dogs, etc.)

What Columns do we need? (age, weight-in-kilograms, weeks, etc.)

Data Cycle: Analyzing with Displays

For each question below, complete the first three steps of the Data Cycle. Once you know what code to write, type it into Pyret and try it out!

Ask Questions	*How many of each species are at the shelter?* *What question do you have?* What type of question is this? (circle one) Lookup Arithmetic Statistical
Consider Data	*What Rows should we investigate? (All the rows, just the cats, fixed dogs, etc.)* *What Columns do we need? (age, weight-in-kilograms, weeks, etc.)*
Analyze Data	

Ask Questions	*How many of each sex are at the shelter?* *What question do you have?* What type of question is this? (circle one) Lookup Arithmetic Statistical
Consider Data	*What Rows should we investigate? (All the rows, just the cats, fixed dogs, etc.)* *What Columns do we need? (age, weight-in-kilograms, weeks, etc.)*
Analyze Data	

Ask Questions	*What question do you have?* What type of question is this? (circle one) Lookup Arithmetic Statistical
Consider Data	*What Rows should we investigate? (All the rows, just the cats, fixed dogs, etc.)* *What Columns do we need? (age, weight-in-kilograms, weeks, etc.)*
Analyze Data	

Question Types: Animals

A subset of the whole Animals Dataset is shown in the table below.

name	species	sex	age	fixed	legs	pounds	weeks
Sasha	cat	female	1	false	4	6.5	3
Sunflower	cat	female	5	true	4	8.1	6
Felix	cat	male	16	true	4	9.2	5
Sheba	cat	female	7	true	4	8.4	6
Billie	snail	hermaphrodite	0.5	false	0	0.1	3
Snowcone	cat	female	2	true	4	6.5	5
Wade	cat	male	1	false	4	3.2	1
Hercules	cat	male	3	false	4	13.4	2
Toggle	dog	female	3	true	4	48	1

Using this table - or the full dataset - write three questions of each type below.

	Question	Type
1		Lookup
2		Lookup
3		Lookup
4		Arithmetic
5		Arithmetic
6		Arithmetic
7		Statistical
8		Statistical
9		Statistical

Using Transformers: Sort, Filter, Transform Attribute, Build Attribute

Transformers allow us to transform datasets to produce new, distinct output datasets, instead of modifying the original input dataset itself. We use them to manipulate tables and enable low-stakes "what if?" exploration.

- In this course, we'll be using a variety of Transformers.
  - `Sort` - We must provide `Sort` with a dataset, a sort method (we'll focus on sorting by attribute in this course), and a sort direction. `Sort` produces a copy of the input dataset, with cases sorted by the selected attribute.
  - `Filter` - We must provide `Filter` with an expression that evaluates to either true or false. `Filter` then produces a copy of the input dataset that only has the cases for which the formula was true.
  - `Transform Attribute` - We provide `Transform Attribute` with an attribute to transform, a new name for the transformed attribute, and an expression to apply to each row. The `Transform Attribute` replaces the value of the designated attribute with the result of the expression. `Transform Attribute` produces a copy of the input dataset with the designated row transformed.
  - `Build Attribute` - We provide an expression that is used to calculate an attribute's new values. `Build Attribute` produces a copy of the input dataset with the new attribute added.
- With the exception of `Sort`, each of the above Transformers is attached to a unique expression. It's important to get the expression just right, or the Transformer will produce an error. Strings belong inside quotation marks, but Booleans do not! Attribute names that include units (for instance, `Age (years)`) belong inside tick marks (`). }

Sort Transformer

Make sure you're logged into CODAP with the Animals Dataset open. Open the Transformers plugin, and select `Sort`.

Creating a Sort Transformer

- Select `Sort` from the drop-down menu of Transformers.
- We want to give each Transformer a useful name, one which clearly describes what that Transformer does. Generally, a brief hyphenated description works - for instance: `order-by-pounds`.

1) What would be a good name for a Transformer that organizes the table alphabetically by animals' names?

_____

- Type in the name you wrote above.

2) Which dataset do you want CODAP to apply this Transformer to? _____

- Select the appropriate dataset from the drop-down menu.
- Select `Name` as the attribute to sort.

3) Which direction will you select? Why?

_____

Applying and Saving the Sort Transformer

4) Apply the Transformer. What happens?

_____

5) What do you notice about the name of the new dataset?

_____

_____

_____

6) What happens when you save a Transformer?

_____

_____

_____

7) How is saving a Transformer different from saving a dataset?

_____

_____

_____

Filter

Make sure you're logged into the Animals Starter File in CODAP.

Creating a "Filter" Transformer

Open the Transformers plugin, and select `Filter`.

1) Enter a `filter-is-fixed` as the name. What does this name tell you about the Transformer you are creating?

2) Which dataset should we select, to apply this Transformer?

3) Examine the contract. Why does the provided contract make sense?

4) Write a purpose statement.

- Enter `Fixed = true` as the expression.

5) Select `Apply Transformer`. What happens?

6) Try typing `fixed = true` as the expression (instead of `Fixed = true`). What happens?

7) What happens when you enter `FIXED = true`? How about `Fixed = TRUE`?

More Filtering

8) Create, save, and apply a Transformer called `filter-is-old` that creates a new dataset with animals older than 5 years. Note that `Age (years)` needs to be typed within backticks``` in order for CODAP to recognize it as an attribute. (The backtick key is in the upper left corner of the keyboard.) How many rows does the resulting table have?

9) Create, save, and apply a Transformer called `filter-is-dog` that creates a new dataset with only dogs. How many dogs are there at the shelter?

Transform

Make sure you're logged into CODAP with the Animals Dataset open.

Creating a "Transform" Transformer

- Select `Transform Attribute` in the Transformer plugin.
- Enter `transform-is-young` as the name. This Transformer will replace all ages less than 5 with the Boolean `true`. In other words, it will *transform* our "age" column into a column that tells us if an animal is young or not.
- Select the Animals Dataset.

1) What would be an appropriate name for our transformed attribute? _____

2) Examine the syntax for the expression Domain (row) is provided

3) Let's write a purpose statement. What do you think we want a Transformer called `transform-is-young` to do?

_____

- Enter `Age (years) < 5` as the expression. (Include tick marks around `Age (years)`.)
- Apply the Transformer. To define the Transformer for future use, select `Save`.

More Transforming

4) Now, try creating a Transformer called `weight-in-kg`. (Note: To convert pounds to kilograms, divide pounds by 2.205.) How many kilograms is the heaviest animal in the shelter?

_____

Build Attribute

Make sure you're logged into CODAP with the Animals Dataset open.

Creating a Build Attribute Transformer

1) Now select `Build Attribute` in the Transformer plugin.

2) Enter `build-is-heavy` as the name. What does this name tell you about the Transformer we are creating?

---

3) Select `animals-dataset`.

4) What would be an appropriate name for our *new* attribute?

---

5) Examine the contract. A domain is provided (row), but not a range. What is the desired output for `build-is-heavy`?

---

6) Let's write a purpose statement. What do you think we want a Transformer called `build-is-heavy` to do?

---

7) Enter `Weight > 40` as the expression.

8) Apply the Transformer. To define the Transformer for future use, select `Save`.

More Building

9) Now, try creating a Transformer called `updated-age`. This Transformer will give the animals' ages one year from today. How many animals are 9 years-old one year from today?

Expression Cards

Print and cut these out, for use with the `Filter` Transformer activity in this lesson. You can make up your own using the blanks provided at the bottom!

Do they wear glasses? `Student -> Boolean` consumes a student from the dataset, produces true if they are wearing glasses	**Are they taller than you?** `Student -> Boolean` consumes a student from the dataset, produces true if they are taller than you
Do they have long hair? `Student -> Boolean` consumes a student from the dataset, and produces true if they have long hair	**Do they have red hair?** `Student -> Boolean` consumes a student from the dataset, and produces true if they have red hair
Are they shorter than you? `Student -> Boolean` consumes a student from the dataset, and produces true if they are shorter than you	**Are they smiling?** `Student -> Boolean` consumes a student from the dataset, and produces true if they are smiling
Does their name begin with a vowel? `Student -> Boolean` consumes a student from the dataset, and produces true if the first letter of their name is a vowel	**Are they wearing jeans?** `Student -> Boolean` consumes a student from the dataset, and produces true if they are wearing jeans

What Table Do We Get?

The table below represents four animals from the shelter:

name	sex	age	fixed	species	pounds
`"Toggle"`	`"female"`	`12`	`true`	`"dog"`	`48`
`"Fritz"`	`"male"`	`4`	`false`	`"dog"`	`92`
`"Nori"`	`"female"`	`6`	`true`	`"dog"`	`35.3`
`"Sunflower"`	`"female"`	`2`	`false`	`"cat"`	`51.6`

Below is a list of saved Transformers. What table do we get, after applying each one? *Match* each expressionto the table we get .

`order-by-age`
Transformer: Sort
Attribute to Sort By: age
Direction: ascending

1

`order-by-weight`
Transformer: Sort
Attribute to Sort By: pounds
Direction: descending

2

`filter-is-old`
Transformer: Filter
Formula to Filter By: Age > 10

3

`filter-is-fixed`
Transformer: Filter
Formula to Filter By: `Fixed = true`

4

`filter-is-dog`
Transformer: Filter
Formula to Filter By: `species = "dog"`

5

`build-is-dog`
Transformer: Build Attribute
Formula for New Attribute Values: `species = "dog"`

6

`build-is-fixed`
Transformer: Filter
Formula to Filter By: `fixed = true`

7

A Produces a table with Toggle, Fritz, and Nori - but not Sunflower.

B Produces a table of all four animals, sorted youngest-to-oldest

C Produces a table, with only Toggle.

D Produces an identical table with an extra column called "dog", whose values are true, true, true, false

E Produces a table containing only Nori and Toggle.

F Produces a table with all four animals, sorted from heaviest to lightest.

G Won't run: will produce an error

Transformer Chaining

Transformer chaining allows us to apply multiple Transformers with fewer clicks.

Let's say we create the following Transformers:

- `build-in-kgs` - builds a new column with pounds converted to kilograms
- `filter-is-lightweight` - keeps all rows where weight is no more than 9 kilograms
- `sort-ascending` - sorts the dataset from lightest to heaviest weight

When we *save* Transformers, we can apply them to new datasets - preventing lots of duplicate work. Additionally, when we have a chain of transformed datasets and change the original dataset, the updates will flow through the chain.

Order Matters! Transformers must be applied in the appropriate order given the dataset. For example, trying to order a table by a column that hasn't been built will result in an error.

Design Recipe

Directions : Define a Transformer called `filter-is-dog` , whose expression consumes a `Row` of the animals table and computes whether the animal is a dog.

Transformer (check one) ☐ **Filter** ☐ **Transform** ☐ **Build**

_____
Transformer name

Example Tables ☐

What gets filtered/transformed/built? In the sample tables below, (if needed) add the relevant columns.

Original Table

name	species

Transformed Table

name	species

Contents (Contract, Purpose Statement, and Expression) ☐

_____ -> _____
Domain *Range*

_____
Purpose: what does the formula do for each row?

_____
i.e. Weight < 20 or Species = "rabbit". Pay careful attention to capitalization and quotation marks.

Design Recipe 2

Directions: Create and save a Transformer that produces a copy of the animals table with only animals that are more than 12 years old.

Transformer (check one) ☐ **Filter** ☐ **Transform** ☐ **Build**

Transformer name

Example Tables

What gets filtered/transformed/built? In the sample tables below, (if needed) add the relevant columns.

Original Table

name	age (years)

Transformed Table

name	age (years)

Contents (Contract, Purpose Statement, and Expression)

Domain -> *Range*

Purpose: what does the formula do for each row?

i.e. Weight < 20 or Species = "rabbit". Pay careful attention to capitalization and quotation marks.

Directions: Create and save a Transformer which returns true in a *new column* if an animal's name contains the letter `s`.

Transformer (check one) ☐ **Filter** ☐ **Transform** ☐ **Build**

Transformer name

Example Tables

What gets filtered/transformed/built? In the sample tables below, (if needed) add the relevant columns.

Original Table

name	species

Transformed Table

name	species	S name

Contents (Contract, Purpose Statement, and Expression)

Domain -> *Range*

Purpose: what does the formula do for each row?

i.e. Weight < 20 or Species = "rabbit". Pay careful attention to capitalization and quotation marks.

Chaining

You have created and saved the following Transformers. (Read them *carefully!*)

`filter-is-fixed`	`filter-is-young`	`order-by-sex`
Transformer: Filter Expression: Fixed = true	Transformer: Filter Expression: Age < 4	Transformer: Sort Attribute: Sex Direction: descending
`order-by-age`	**`order-by-pounds`**	**`order-by-sex`**
Transformer: Sort Attribute: Age Direction: ascending	Transformer: Sort Attribute: Pounds Direction: descending	Transformer: Sort Attribute: Sex Direction: descending

The table below represents four animals from the shelter:

name	sex	age	fixed	pounds
`"Toggle"`	`"female"`	`3`	`true`	`48`
`"Fritz"`	`"male"`	`4`	`true`	`92`
`"Nori"`	`"female"`	`6`	`true`	`35.3`
`"Maple"`	`"female"`	`3`	`true`	`51.6`

What table do we when we apply the indicated Transformers? Match the instructions on the left with the outcome on the right.

Apply `order-by-age`	**1**	**A**	Produces a table containing only Toggle and Maple
Apply `filter-is-fixed`	**2**	**B**	Produces a table of only young, fixed animals
Apply `filter-is-young`	**3**	**C**	Produces a table, sorted youngest to oldest
First, apply `filter-is-young` Next, apply `filter-is-fixed`	**4**	**D**	Produces a table containing Maple and Toggle, in that order
First, apply `filter-is-young` Next, apply `order-by-pounds`	**5**	**E**	Produces a table containing the same four animals
Apply `order-by-pounds`	**6**	**F**	Produces a table of all four animals, with Fritz first

Chaining 2: Order Matters

You have created and saved the following Transformers. (Read them *carefully!*)

`build-kilos`	`order-by-kilos`	`filter-is-female`
Transformer: Build Attribute Name of new attribute: `kilos` Expression: pounds/2.2	Transformer: Sort Attribute: `kilos` Direction: ascending	Transformer: Filter Expression: `sex = "female"`
`filter-is-heavy`	**`order-by-sex`**	**`build-is-female`**
Transformer: Filter Expression: `kilos > 25`	Transformer: Sort Attribute: `sex` Direction: ascending	Transformer: Build Attribute Name of new attribute: `is-female` Expression: `sex = "female"`

The table below represents four animals from the shelter:

name	sex	age	fixed	pounds
`"Toggle"`	`"female"`	3	`true`	48
`"Fritz"`	`"male"`	4	`true`	92
`"Nori"`	`"female"`	6	`true`	35.3
`"Maple"`	`"female"`	3	`true`	51.6

Match the CODAP instructions (left) to what they do (right). **Note: one description might match multiple expressions!**

Apply `order-by-kilos`	1	A	Produces a table containing Toggle, Nori and Maple, with an extra column showing their weight in kilograms
Apply `filter-is-female` Next, apply `build-kilos`	2	B	Produces a table containing Maple, Nori and Toggle (in that order)
Apply `build-kilos` Next, apply `filter-is-heavy`	3	C	Produces a table containing only Fritz.
First, apply `filter-is-heavy` Next, apply `build-kilos`	4	D	Won't run: will produce an error
First, apply `build-kilos` Next, apply `filter-is-heavy` Next, apply `order-by-sex`	5	E	Produces a table containing only Fritz, with two extra columns.
Apply `build-is-female` Apply `build-kilos` Apply `filter-is-heavy`	6	F	Produces a table containing Maple and Fritz

What's on your mind?

Table Transformations with Chaining

You have created and saved the following Transformers. (Read them *carefully!*)

`filter-is-fixed`	`filter-is-young`	`filter-is-female`	`build-kilos`
Transformer: Filter Expression: Fixed = true	Transformer: Filter Expression: Age < 4	Transformer: Filter Expression: Sex = “female”	Transformer: Build Attribute Name of new attribute: Kilos Expression: Weight (lbs) / 2.2
`filter-is-heavy`	`filter-is-dog`	`filter-is-cat`	
Transformer: Filter Expression: Kilos > 25	Transformer: Filter Expression: Species = “dog”	Transformer: Filter Expression: Species = "cat"	

Solve the word problems below using the Transformers listed above.

	Prompt	Does order matter?	List the transformers in order.	Use function notation to predict the table’s name.
1	Create a table of all female cats.	No	Apply `filter-is-female` . Apply `filter-is-cat` .	filter-is-cat(filter-is-female(animals-table))
2	Create a table of all young, fixed animals.			
3	Create a table of all female dogs, converting their weight to kilograms.			
4	Create a table of all heavy dogs.			
5	Create a table of all young, fixed dogs, sorted from lightest to heaviest.			

Probability, Inference, and Sample Size

How can you tell if a coin is fair, or designed to cheat you? Statisticians know that a fair coin should turn up "heads" about as often as "tails", so they begin with the **null hypothesis:** they assume the coin is fair, and start flipping it over and over to record the results.

A coin that comes up "heads" three times in a row could still be fair! The odds are 1-in-8, so it's totally possible that the null hypothesis is still true. But what if it comes up "heads" five times in a row? Ten times?

Eventually, the chances of the coin being fair get smaller and smaller, and a Data Scientist can say "this coin is a cheat! The chances of it being fair are one in a million!"

By sampling the flips of a coin, we can *infer* whether the coin itself is fair or not. **Inference** plays a major role in Data Science and Statistics!

- If we survey pet owners about whether they prefer cats or dogs, the **null hypothesis** is that the odds of someone preferring dogs are about the same as them preferring cats. And if the first three people we ask vote for dogs (a 1-in-8 chance), the null hypothesis could still be true! But after five people? Ten?
- If we're looking for gender bias in hiring, we might start with the null hypothesis that no such bias exists. If the first three people hired are all men, that doesn't necessarily mean there's a bias! But if 30 out of 35 hires are male, this is evidence that undermines the null hypothesis and suggests a real problem.
- If we poll voters for the next election, the **null hypothesis** is that the odds of voting for one candidate are the same as voting for the other. But if 80 out of 100 people say they'll vote for the same candidate, we might reject the null hypothesis and infer that the population as a whole is biased towards that candidate!

Sample size matters! The more bias there is, the smaller the sample we need to detect it. Major biases might need only a small sample, but subtle ones might need a huge sample to be found. However, choosing a **good sample** can be tricky!

Random Samples are a subset of a population in which each member of the subset has an equal chance of being chosen. A random sample is intended to be a representative subset of the population. The larger the random sample, the more closely it will represent the population and the better our inferences about the population will tend to be.

Grouped Samples are a subset of a population in which each member of the subset was chosen for a specific reason. For example, we might want to look at the difference in trends between two groups ("Is the age of a dog a bigger factor in adoption time v. the age of a cat?"). This would require making grouped samples of *just the dogs* and *just the cats*.

Finding the Trick Coin

There are three coins defined in the [Fair Coins Starter File](): `coin1`, `coin2`, and `coin3`.

One of these coins is fair, one will land on "heads" 75% of the time, and one will land on "heads" 90% of the time. ***Which one is which?***

1) In the table below, record five flips of each coin and the number of "heads" you saw. Then convert that number into a *percentage*. Finally, write a "Y" or "N" depending on whether you think each coin is fair based on your sample.

Sample	coin1	coin2	coin3
1			
2			
3			
4			
5			
#heads	/5	/5	/5
% heads	%	%	%
fair?			

2) In the table below, record 15 more flips of each coin and the number of "heads" you saw. Then convert that number into a *percentage*. Finally, write a "Y" or "N" depending on whether you think each coin is fair based on your sample.

Sample	coin1	coin2	coin3
6			
7			
8			
9			
10			
11			
12			
13			
14			
15			
16			
17			
18			
19			
20			
#heads	/20	/20	/20
% heads	%	%	%
fair?			

3) Which weighted coin did you identify first: fair, 75%, or 90%? _____

4) Why was that coin the easiest to find? _____

_____

Sampling and Inference

1) In the screenshots of the "Sampler" (below), show how you would create a small random sample of 10 animals and a large random sample of 40 animals. *To create two separate tables (rather than a single hierarchical table), re-select and re-open "Sampler" from the Plugins menu before each sampling simulation.*

2) In the options tab, did you select "with replacement" or "without replacement"? Why?

_____

3) Make a bar chart for the animals in each sample, showing percentages of fixed and unfixed.

- The percentage of fixed animals in the entire population is: **47.7%**
- The percentage of fixed animals in the small sample is: _____
- The percentage of fixed animals in the large sample is: _____

4) Make a bar chart for the animals in each sample, showing percentages for each species.

- The percentage of tarantulas in the entire population is: **roughly 5%**
- The percentage of tarantulas in the small sample is: _____
- The percentage of tarantulas in the large sample is: _____

5) Direct the sampler to generate a different set of random samples of these sizes. Make a new bar chart for each sample, showing percentages for each species.

- The percentage of tarantulas in the entire population is: **roughly 5%**
- The percentage of tarantulas in the small sample is: _____
- The percentage of tarantulas in the large sample is: _____

6) Which repeated sample gave us a more accurate inference about the whole population? Why?

_____

_____

Predictions from Samples

1) Use the Sampler plugin to create the following tables and give them the specified names. (Re-open Sampler *before* creating each table to avoid getting one massive hierarchical table!)

- 10 animals randomly selected without replacement ⇒ tiny-sample
- 20 animals randomly selected without replacement ⇒ small-sample
- 40 animals randomly selected without replacement ⇒ medium-sample
- 80 animals randomly selected without replacement ⇒ large-sample

2) Make a bar chart of the species in the tiny sample.

- What animals are in the sample? _____
- Create a new random tiny-sample. Create another bar-chart of species in the tiny-sample. What animals are in the sample? _____
- Create another random sample and make another bar-chart of species in the tiny-sample. Based on these samples, how many species of animals do you think are at the shelter? _____
- Which species do you think there are the most of at the shelter? _____

3) What did you learn from taking multiple samples that you wouldn't have known if you'd only taken a single sample?

_____

4) Now use the small sample to make a bar chart of the species.

- What animals are in the sample? _____
- Create a new random sample and make another bar chart of species in the small sample. What animals are in the sample? _____

5) Now that you've seen the small sample, how has your sense of the distribution of the species changed?

_____

6) Now use the medium sample to make a bar chart of the species. If there are about 400 animals at the shelter, how many of each species would you predict there to be?

_____

7) Now use the large to make a bar chart of the species. If there's anything you'd like to change about your prediction now that you've seen the large sample, record it here.

_____

8) Let's see how accurate your prediction is… When you're ready, make a bar chart of Animals Table 2.

- Which predictions were closest? _____
- Which predictions were off? _____
- Were there any surprises? _____

9) In the real world, we usually don't have access to a whole dataset to check predictions against! How could we test…

- *Every giraffe on the planet?*
- *Everyone who has ever come in contact with a covid-positive person?*
- *Every person who identifies as queer?*

What strategies can we use to make sure that predictions from samples are as close to accurate as possible?

_____

_____

Grouped Samples from the Animals Dataset

You've already created and saved the following transformers: `filter-is-old`, `filter-is-young`, `filter-is-cat`, `filter-is-dog`, `filter-is-female`, `filter-is-fixed`, and `filter-has-s-name`. Provide the transformers you would use in the order you would use them. We've given you the solution for the first sample, to get you started.

	Subset	List the transformers *in order*	Use function notation
1	Kittens	`filter-is-young`, `filter-is-cat`	filter-is-young(filter-is-cat(animals-table))
2	Puppies		
3	Fixed Cats		
4	Cats with "s" in their name		
5	Old Dogs		
6	Fixed Animals		
7	Old Female Cats		
8	Fixed Kittens		
9	Fixed Female Dogs		
10	Old Fixed Female Cats		

Displaying Data

Fill in the tables below, then use CODAP to make the following displays. The first table has been filled in for you.

1) A bar chart showing how many puppies are fixed or not.

What Rows?	Which Column(s)?	What Display?
puppies	*fixed*	*bar-chart*

2) A bar chart showing how many heavy dogs are fixed or not.

What Rows?	Which Column(s)?	What Display?

3) A box plot of the number of pounds that kittens weigh.

What Rows?	Which Column(s)?	What Display?

4) Describe **your own grouped sample** here, and fill in the table below.

_____

What Rows?	Which Column(s)?	What Display?

Data Cycle: Analyzing Categorical Data

Ask Questions

How many of each species are fixed at the shelter?

What question do you have?

What type of question is this? (circle one) Lookup Arithmetic Statistical

Consider Data

What Rows should we investigate? (All the rows, just the cats, fixed dogs, etc.)

What Columns do we need? (age, weight-in-kilograms, weeks, etc.)

Analyze Data

If you only need some rows, write an expression for your Filter Transformer here.

If you need to Transform or Build an attribute, write the expression for your Transformer here.

Interpret Data

What did you find out? What can you infer?

What new question(s) does this raise?

Ask Questions

Are there more female cats than male cats at the shelter?

What question do you have?

What type of question is this? (circle one) Lookup Arithmetic Statistical

Consider Data

What Rows should we investigate? (All the rows, just the cats, fixed dogs, etc.)

What Columns do we need? (age, weight-in-kilograms, weeks, etc.)

Analyze Data

If you only need some rows, write an expression for your Filter Transformer here.

If you need to Transform or Build an attribute, write the expression for your Transformer here.

Interpret Data

What did you find out? What can you infer?

What new question(s) does this raise?

What's on your mind?

Design Recipe

Directions:

Transformer (check one) ☐ **Filter** ☐ **Transform** ☐ **Build**

Transformer name

Example Tables ☐

What gets filtered/transformed/built? In the sample tables below, (if needed) add the relevant columns.

Original Table	

Transformed Table	

Contents (Contract, Purpose Statement, and Expression) ☐

->

Domain *Range*

Purpose: what does the formula do for each row?

i.e. Weight < 20 or Species = "rabbit". Pay careful attention to capitalization and quotation marks.

Directions:

Transformer (check one) ☐ **Filter** ☐ **Transform** ☐ **Build**

Transformer name

Example Tables ☐

What gets filtered/transformed/built? In the sample tables below, (if needed) add the relevant columns.

Original Table	

Transformed Table	

Contents (Contract, Purpose Statement, and Expression) ☐

->

Domain *Range*

Purpose: what does the formula do for each row?

i.e. Weight < 20 or Species = "rabbit". Pay careful attention to capitalization and quotation marks.

Design Recipe

Directions:

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